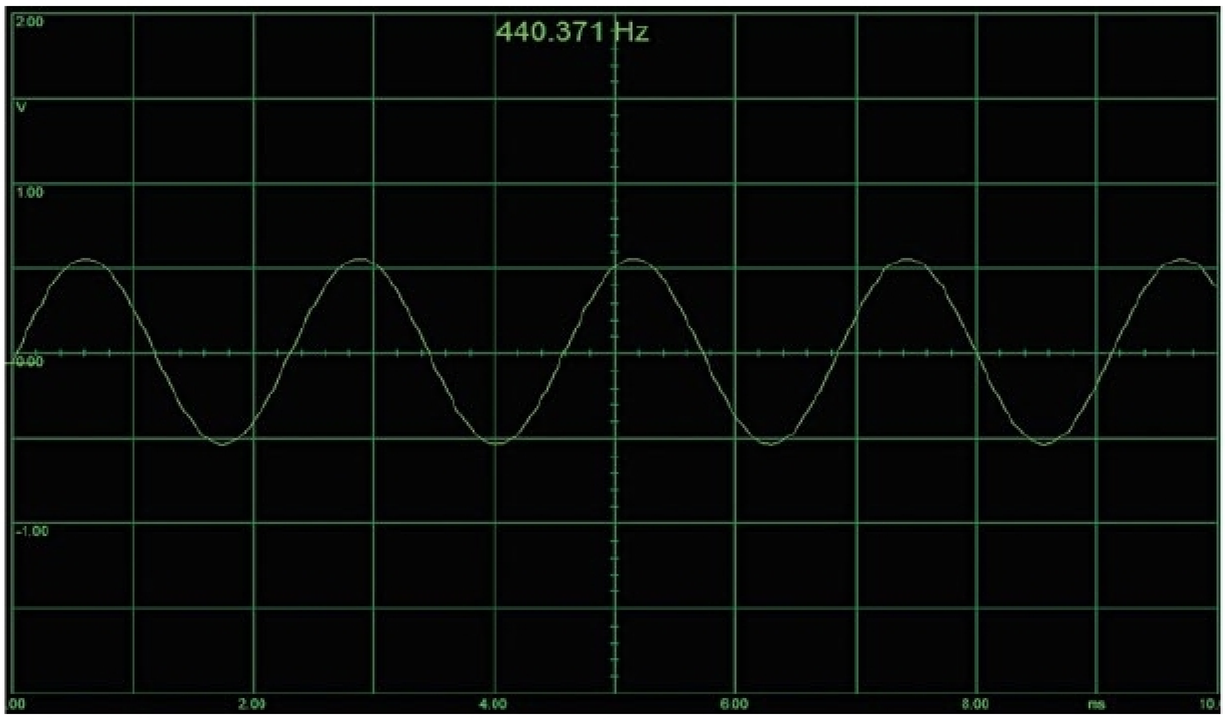

Fig. 12: Frequency captured on Zelscope software

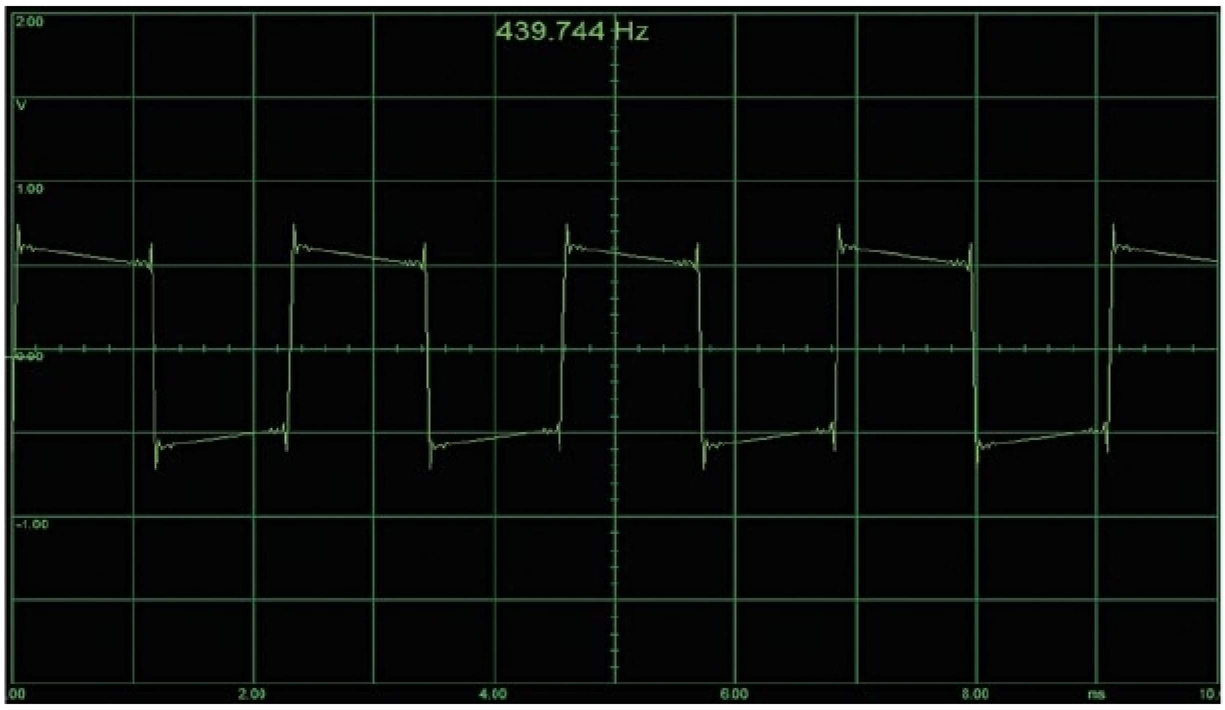

Fig. 13: Square-wave signal obtained on Zelscope software

text values are also available in this software.

Connect the audio jack from the cellphone output port to the microphone port of the laptop/PC. If waveform is not obtained on Zelscope software, adjust the microphone settings on the PC.

Analysis of 440Hz sinewave signal. Output from the 3.5mm jack of Android smartphone is given to the PC through a USB sound card. A 440Hz sinewave signal generated on Android app, which is to be analysed on Zelscope, is shown in Fig. 11.

The 440Hz sinewave signal tested on Zelscope software is shown in Fig. 12.

Analysis of 440Hz square-wave signal. Similarly, a 440Hz square-wave signal generated using the Android App and its corresponding frequency on Zelscope software is shown in Fig. 13.

Noise generated on the PC-based software is less than that observed on the DSO. It can be minimised using a good audio source and good-quality cable.

Caution. Use a proper protection circuit while connecting Android smartphone to the PC for Function Generator application. It is recommended to use a USB sound card instead of directly connecting the smartphone audio output to mic/audio input of the PC or laptop. EFY

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